

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	19-February-2026	Section	6A , 6B
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. Consider an **HTTP/2** connection downloading 3 objects: Object A (three frames, i.e. A1, A2, A3), Object B (one frame i.e. B1) and Object C (one frame i.e. C1). Each frame takes 10 ms to transmit.
[2 + 3 = 5 Marks] [CLO 1]

a) List the sequence of frame arrival at the client if Round Robin (FCFS with preemption) scheduling is used to prevent Head-of-Line (HOL) blocking.

b) Contrast this with HTTP/1.1 where Object A must be fully downloaded before B and C. How much earlier does Object B finish in HTTP/2 compared to HTTP/1.1? (i.e. calculate the time taken for Object B to be fully downloaded in HTTP/1.1 compared to in HTTP/2.)

Q2. We notice an increase in the total delay (time) for those requests that encounter a web cache (a penalty because of the involvement of the cache). Let L be the average LAN delay, A be the access link delay, I be the Internet delay, and R be the total response time. Also assume that P is the average cache penalty delay. Given that $L = 10$ ms, $I = 180$ ms, $P = 40$ ms, and $A = 60$ ms. **[5 + 5 = 10 Marks] [CLO 1]**

(i) Find the average response delay (time), R , when no cache is used?

(ii) Find the average response delay (time) when a cache is used and the hit ratio is 80%?